

Fundamentals of Excel Calculations

All formulas in Excel must begin with an **equals (=) sign**. Excel uses specific symbols to perform operations:

- **Arithmetic Symbols:**
 - **+**: Addition.
 - **-**: Subtraction.
 - *****: Multiplication.
 - **/**: Division.
 - **()**: Parentheses (used to group parts of a formula).
- **Comparison Symbols:**
 - **>**: Greater than.
 - **<**: Less than.
 - **<>**: Not equal to.
 - **=**: Equal to.
 - **>=**: Greater than or equal to.
 - **<=**: Less than or equal to.

Basic Excel Functions

Functions are pre-built formulas that perform specific tasks. They can be added to a cell through various methods, including the **Auto sum** feature.

Statistical Functions

- **SUM()**: Adds together a range of numerical values.
- **COUNT()**: Counts the number of cells that contain numbers.
- **AVERAGE()**: Calculates the arithmetic mean of a group of values.
- **MAX()**: Identifies the highest value in a set.
- **MIN()**: Identifies the lowest value in a set.

Rounding and Mathematical Functions

- **ROUND()**: Used to increase or decrease decimal values to a specified number of digits.
 - *Syntax*: `ROUND(Value, num of digits)`.
- **ROUNDUP()**: Always rounds a number away from zero.
- **ROUNDDOWN()**: Always rounds a number toward zero.
- **FLOOR()**: Rounds a number down to the **previous nearest** specified multiple (such as 10, 50, or 100).
- **CEILING()**: Rounds a number up to the **next nearest** specified multiple (such as 10, 50, or 100).

